

Nettem Praneetha

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Education

Velagapudi Ramakrishna Siddhartha Engineering College, Vijayawada <i>B.Tech in Information Technology, CGPA – 8.52</i>	2022 – 2026
Sri Gayathri Junior College, Vijayawada <i>Intermediate, Percentage – 88.2</i>	2020 – 2022
Sri Chaitanya High School, Jaggaiahpet <i>Class X SSC, Percentage – 98.1</i>	2020

Skills

Programming Languages :	Python, HTML, CSS
Databases :	MySQL, Firebase
CS Fundamentals :	OOPS, Database Management System, DSA (Basic)
Soft Skills :	Problem Solving, Time Management, Team Collaboration
Tools / Platforms:	Git, IoT, MS Word, MS Excel

Experience

Eisystems Services Intern

Training on Machine Learning with Python

- Developed a machine learning model to predict the likelihood of heart disease using clinical patient data with preprocessing, training, and evaluation techniques.
- **Tech Used:** Python, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn

Research Paper

Real-Time Fire Detection and Irrigation Management System using IoT and Image Processing (Published) 2025

- Authored research paper, "Real-Time Fire Detection and Irrigation Management System using IoT and Image Processing," presented at 19th INDIACOM - 2025 (IEEE Conference ID: 66777)

Projects

YOLOv8 and ESP32-Powered System for Agricultural Fire Prevention (2024)

- Developed an integrated IoT and computer vision-based agricultural wildfire detection system using ESP32 environmental sensors and a custom-trained YOLOv8 model, achieving 87.5% accuracy
- Implemented an automated sprinkler response and real-time monitoring through a Firebase-connected mobile application for smart irrigation and rapid fire alerts.
- **Tech Used:** Python, YOLOv8, ESP32, IoT Sensors, Firebase, Mobile App Development.

Soil Fertility Mapping and Crop Recommendation System using GIS and Remote Sensing (2025)

- Developed and compared two integrated frameworks (IDW–XGBoost and Kriging–TabNet) for soil fertility mapping and crop recommendation in Guntur district.
- Generated GIS-based spatial fertility maps using soil nutrients (N, P, K, pH, OC, EC) and climatic parameters for precision agriculture.
- Achieved 99.09% accuracy using the TabNet deep learning model for location-specific crop prediction.
- **Tech Used:** Python, GIS, Remote Sensing, XGBoost, TabNet, Kriging, IDW, Scikit-learn, Google Earth Engine.

Certifications

- AWS Certified Cloud Practitioner – Amazon Web Services (2025)
- DBMS and MySQL (Udemy, Apr 2024)
- The Joy of Computing using Python (NPTEL, Jan–Apr 2024)

Co-curricular Activities

- Member of CSI Student Chapter, contributing to technical and cultural initiatives.
- Organized and coordinated the technical and cultural event “Splash” at college, guiding juniors and managing execution.
- Served as Event Coordinator, leading student teams and ensuring smooth conduct of the program.